Mogomane Digashu_26304826_SS3_PartD_Github questions

Responses to Client Questions

Question 1: If you could request additional client data to improve your model, what would it be and why?

To assist the model in learning a pattern (i.e., an association) that is related to the minority class (e.g. the returned items), I would like more examples of returned orders. Instead of having an equal number of returned and non-returned orders within the real-world dataset, I would investigate methods to correct the imbalance during the model's training process; one method is by employing class weighting, another is through resampling.

In addition to product fit data, which would include standardized measurements of garments, customer sizing/fit preference information, images of products, material characteristics, etc. I believe we can use the following to further enhance the ability of XGBoost to find more relevant relationships between variables about returns. For instance, customers may return clothing/shoes due to an issue with sizing, or simply because the product was different than what they expected.

Once you combine all your additional data with your prior order history/product detail, XGBoost will then be able to better identify patterns associated with returns. The improved identification of returns will allow you to generate better predictions regarding return risk. Once you have better return-risk predictions, you can provide your customers better-size recommendations, as well as more accurate product descriptions. This may lead to fewer returns, lower reverse logistics cost, and less of an impact on our environment.

One important thing that we need to take into consideration is that all the predictive features must be available at/before checking out. Return reason(s) may help us determine why there has been a trend towards returning certain types of orders, however they should never be utilized as predictors for the current order if the historical return reason(s) are recorded after the return has occurred.

Question 2: How would your model perform against a model that predicts "Not Returned" for every order, in terms of metrics?

A baseline model which consistently predicts "Not Returned" will have an overall accuracy greater than mine since most test cases are non-returns. But a baseline will never discover any returned orders. In my unseen test dataset (1,911 orders), 428 returned orders, 1483 did not. Therefore, by forecasting "Not Returned", on each order my baseline model could forecast correctly at 77.6%. And in contrast, my XGBoost model had an accuracy rate of 71.4% and was able to forecast correctly around 61% of all actual returns.

The following table shows the comparison of metrics from both XGBoost and Baseline Models:
XG Boost Baseline Model

Accuracy 71.43% 77.60%
Recall for Returned 60.51% 0.00%
Precision for Returned 40.72% 0.00%
F1-score for Returned 48.68% 0.00%
ROC-AUC 0.7315 0.50

For my project, ROC curve was assumed to represent the baseline model using the same predicted value for every order. This demonstrates how simply evaluating model accuracy can be misleading if the two target classes being modelled are unbalanced. While the Baseline has better accuracy, it cannot identify potential high-cost return orders. Because XGBoost identifies many actual returns, it provides a valuable resource to allow STADIOnlot to explore possible ways to lower return cost.

But like other classification models, there exists a trade-off with XGBoost's precision for returned orders being 40.72%, implying that only around 41% of orders classified as returned by the model are returned. Therefore, there is also a possibility of unnecessary intervention and/or further expense.

Question 3: Your entire test set appears to be from 2026. Why not include 2026 data in the training set?

I used a chronological 70%/15%/15% train-validation-test split to simulate how the model would perform when predicting returns for future orders. The training set contains 8,913 orders dated from September 2022 to 14 June 2025. The validation set contains 1,910 orders dated from 14 June 2025 to 25 January 2026, while the test set contains 1,911 orders dated from 25 January to 30 August 2026.

I deliberately kept the later test data separate so that the model could be evaluated on orders it had not seen during training or model development. This provides a more realistic assessment of how the model might perform when deployed to predict returns for new orders.

I could include earlier 2026 orders in the training set if I redesigned the chronological split, if I reserved a later period as the final test set. However, I would not include the existing test-set observations in training before evaluating the model, because doing so would compromise the independence of the final evaluation.

Therefore, the purpose of keeping the 2026 test set separate is not to exclude useful training data unnecessarily, but to assess how well the model generalizes to future, unseen orders. This is particularly important for a business application where the model would be used to predict return risks for incoming orders.